

Suspension System



Inspection and Verification

1. Road test.
- Verify the customer concern by carrying out a road test on a smooth road. If any vibrations are apparent, refer to Section 100-04.
2. Inspect tires.
- Check the tire pressure with all normal loads in the vehicle and the tires cold. Refer to the Vehicle Certification (VC) label.
 - Verify that all tires are sized to specification. Refer to the VC label.
 - Inspect the tires for incorrect wear and damage. Install new tires as necessary.
3. Inspect chassis and underbody.
4. Inspect for aftermarket equipment.
- Check for aftermarket changes to the steering, suspension, wheel and tire components (such as competition, heavy duty, etc.). The specifications shown in this manual do not apply to vehicles equipped with aftermarket equipment.

Visual Inspection Chart

Mechanical
• Front wheel bearing(s) • Loose or damaged front or rear suspension components • Loose, damaged or missing suspension fastener(s) • Incorrect spring usage • Damaged or sagging spring(s) • Damaged or leaking shock absorber(s) • Worn or damaged suspension bushing(s) • Loose, worn or damaged steering system components • Damaged axle components

5. If an obvious cause for an observed or reported condition is found, correct the cause (if possible) before proceeding to the next step.
6. If the fault is not visually evident, GO to [Symptom Chart — Suspension System](#) or GO to [Symptom Chart — NVH](#).

Symptom Chart — Suspension System

Symptom Chart — Suspension System		
Condition	Possible Causes	Action

• Incorrect thrust angle (dogtracking)	• Rear suspension components	• INSPECT the rear suspension system. CHECK the rear alignment for the correct thrust angle. REPAIR or INSTALL new suspension components as necessary. REFER to Section 204-02.
• Vehicle drifts/pulls	• Unevenly loaded or overloaded vehicle • Tires/tire pressure • Alignment is not within specification • Brake drag • Steering components	• GO to Pinpoint Test A.
• Front bottoming or riding low	• Worn, damaged or incorrect springs • Worn front shock absorbers	• MEASURE the ride height. REFER to Ride Height Measurement in this section. INSTALL new springs as necessary. Refer to the appropriate section in Group 204 for the procedure. • INSTALL new shock absorbers as necessary. Refer to the appropriate section in Group 204 for the procedure.
• Abnormal/incorrect tire wear	• Incorrect tire pressure (rapid center rib or inner and outer edge wear) • Incorrect tire rotation intervals • High-speed cornering • Excessive front or rear toe (inner or outer edge wear) • Excessive negative or positive camber (inner or outer edge wear) • Front or rear suspension components	• ADJUST the tire pressure. REFER to the Vehicle Certification (VC) label. REFER to Section 204-04 Diagnosis and Testing for further tire wear diagnosis. • REFER to Section 100-03. • REFER to Section 204-04 Diagnosis and Testing for further tire wear diagnosis. • INSPECT the front and rear suspension system. REPAIR or INSTALL new suspension components as necessary. Refer to the appropriate section in Group 204 for the procedure.

Sticky steering, poor returnability	- Binding ball joints - Steering components - Caster out of specification	- REFER to the Ball Joint Inspection Component Test in this section. - INSPECT the steering system. INSTALL new components as necessary. REFER to Section 211-02. - CHECK the wheel alignment. REFER to Camber and Caster Adjustment in this section. ADJUST as necessary.
Steering wheel off-center	- Unequal front toe setting (side-to-side) - Steering components	- CHECK the wheel alignment. REFER to Toe Adjustment — Front in this section. ADJUST as necessary. - INSPECT the steering system. INSTALL new components as necessary. For vehicles equipped with Electronic Power Assist Steering (EPAS) , refer to Section 211-00A. For vehicles equipped with Hydraulic Power Assist Steering (HPAS) , refer to Section 211-00B.
Sway or roll	- Overloaded, unevenly or incorrectly loaded vehicle - Loose wheel nut(s) - Shock absorber(s) - Loose stabilizer bracket-to-frame bolts - Worn stabilizer bar bushings or links - Damaged or broken stabilizer bar - Damaged spring(s)	- NOTIFY the customer of incorrect vehicle loading. - TIGHTEN the wheel nut(s) to specification. REFER to Section 204-04. - INSTALL new shock absorbers as necessary. Refer to the appropriate section in Group 204 for the procedure. - TIGHTEN the bolts to specification. Refer to the appropriate section in Group 204 for the procedure. - INSTALL new stabilizer bar bushings or links as necessary. Refer to the appropriate section in Group 204 for the procedure. - INSTALL a new stabilizer bar as necessary. Refer to the appropriate section in Group 204 for the procedure. - INSTALL new springs as necessary. Refer to the appropriate section in Group 204 for the procedure.
Vehicle leans to one side	- Unevenly loaded or overloaded vehicle - Front or rear suspension	- NOTIFY the customer of incorrect vehicle loading. - MEASURE the ride height. REFER to Ride Height Measurement in this section. If the ride

	components	height is not within specification, INSPECT the front and rear suspension systems. INSTALL new suspension components as necessary. Refer to the appropriate section in Group 204 for the procedure.
Wander	- Overloaded, unevenly or incorrectly loaded vehicle - Ball joint(s) - Loose, worn or damaged front wheel bearing(s) - Loose, worn or damaged suspension component(s) - Loose suspension fasteners - Steering components - Wheel alignment (excessive total front toe out)	- NOTIFY the customer of incorrect vehicle loading. - INSPECT the ball joints. REFER to the Ball Joint Inspection Component Test in this section. - INSPECT the wheel bearings. INSTALL new wheel bearings as necessary. - INSTALL new suspension component(s) as necessary. Refer to the appropriate section in Group 204 for the procedure. - INSPECT the suspension fasteners. TIGHTEN to specification. Refer to the appropriate section in Group 204 for the procedure. - INSPECT the steering system. INSTALL new components as necessary. For vehicles equipped with Electronic Power Assist Steering (EPAS) , refer to Section 211-00A. For vehicles equipped with Hydraulic Power Assist Steering (HPAS) , refer to Section 211-00B. - ADJUST as necessary. REFER to Toe Adjustment — Front in this section.

Symptom Chart — NVH

Symptom Chart — NVH		
Condition	Possible Causes	Action
NOTE: NVH symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to Section 100-04. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to Section 100-04 for the next likely system and continue diagnosis.		
Squeak or grunt — noise from the front or rear suspension, occurs more in cold ambient temperature with or without the presence of moisture. More noticeable over rough roads or when turning	- Front stabilizer bar insulators - Leaf spring bushings	Under these conditions, the noise is acceptable.

• Clunk — noise from the front suspension, occurs in and out of turns	• Loose front suspension	• INSPECT for loose nuts or bolts. TIGHTEN to specification. Refer to the appropriate section in Group 204 for the procedure.
• Noise from front suspension when turning — Crunch, clunk or grunt noise from the front suspension occurring when turning to full left or right	• Steering stops	• Normal condition. No action required.
• Clunk — noise from the rear suspension, occurs when shifting from REVERSE to DRIVE	• Loose rear suspension components	• INSPECT for loose or damaged rear suspension components. REPAIR or INSTALL new components as necessary. REFER to Section 204-02.
• Click or pop — noise from the front suspension. More noticeable over rough roads or over bumps	• Worn or damaged ball joint(s)	• CARRY OUT a ball joint inspection. INSTALL new ball joint(s) or control arm(s) as necessary. REFER to Section 204-00.
• Front suspension noise — a squeak, creak or rattle noise. Occurs mostly over bumps or rough roads	• Front suspension components • Loose or damaged shock absorber(s) or shock absorber bushing(s) • Damaged spring or spring mount(s) • Damaged or worn control/radius arm bushing(s) • Worn or damaged stabilizer bar bushings or link(s)	• INSPECT the front suspension. INSTALL new components as necessary. Refer to the appropriate section in Group 204 for the procedure.
• Rear suspension noise — a squeak, creak or rattle noise. Occurs mostly over bumps or rough roads	• Loose or damaged rear shock absorber(s) or shock absorber bushing(s) • Rear leaf spring tip isolator or bushing • Damaged leaf spring or leaf spring mount(s)	• INSPECT the rear suspension. INSTALL new components as necessary. REFER to Section 204-02.

	- Worn or damaged stabilizer bar bushing(s) or link(s) - Rear leaf spring or bushing contaminated with dirt, sand or mud	NOTE: <i>To avoid the re-occurrence of noise, make sure that all contamination and moisture trapped between the individual leaf spring components have been completely removed.</i> NOTE: <i>Do not apply any lubricant or grease to the leaf springs as this will attract dirt and moisture.</i> PRESSURE WASH the leaf springs between the individual spring leaves and the gaps between the spring bushings, mounting brackets and shackles to remove any trapped debris. USE compressed (shop) air to remove any remaining debris and to completely dry between all of the individual spring components.
Shudder — occurs during acceleration from a slow speed or stop	Incorrect ride height causing incorrect driveline angle	REFER to Section 205-00 for driveline angle diagnosis.
Shimmy	- Loose wheel nut(s) - Loose front suspension fastener(s) - Loose front wheel bearing(s) - Shock absorber(s)	TIGHTEN the nut(s) to specification. REFER to Section 204-04. TIGHTEN the fastener(s) to specification. Refer to the appropriate section in Group 204 for the procedure. INSPECT the front wheel bearing(s). INSTALL new bearing(s) as necessary. Refer to the appropriate section in Group 204 for the procedure. INSTALL new shock absorbers as necessary. Refer to the appropriate section in Group 204 for the procedure.
Shimmy — most noticeable on coast/deceleration. Also hard steering condition	Excessive positive caster	CHECK the wheel alignment. REFER to Camber and Caster Adjustment in this section. ADJUST as necessary.
Rough/harsh ride	Incorrect tire	ADJUST the tire pressure. REFER to

	pressure • Shock absorber(s) • Spring(s) • Loose, worn or damaged suspension component(s)	the VC label. • INSTALL new shock absorbers as necessary. Refer to the appropriate section in Group 204 for the procedure. • MEASURE the ride height. REFER to Ride Height Measurement in this section. INSTALL new springs as necessary. Refer to the appropriate section in Group 204 for the procedure. • INSTALL new suspension component(s) as necessary. Refer to the appropriate section in Group 204 for the procedure.
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Pinpoint Tests

Pinpoint Test A: Vehicle Drifts/Pulls

This pinpoint test is intended to diagnose the following:

- Unevenly loaded vehicle
- Tire pressure
- Tire forces
- Brake drag
- Incorrect vehicle alignment
- Steering system

PINPOINT TEST A : VEHICLE DRIFTS/PULLS

A1 CHECK FOR UNEVENLY LOADED VEHICLE	
• Visually check the vehicle for an uneven loading condition.	
Is the vehicle unevenly loaded?	
Yes	ADVISE the customer of uneven loading condition.
No	GO to A2 .
A2 CHECK THE TIRE PRESSURES AND TIRE CONDITIONS	
• Check the tire pressures. Refer to the Vehicle Certification (VC) label located on the driver door jamb. Check the tires for uneven/abnormal wear. Refer to Diagnosis and Testing — Wheels and Tires in Section 204-04.	
Are the tire pressures and tire conditions OK?	
Yes	GO to A3 .
No	ADJUST the tire pressures to the specified pressure or INSTALL new tires as necessary.

A3 ISOLATE TIRE DRIFT/PULL CONDITION

- Cross the front wheel and tire assemblies from left-to-right. Refer to Section 204-04.

Does the vehicle drift/pull?

Yes	If the vehicle drifts/pulls in the opposite direction, tire forces are causing the drift/pull. ROTATE the wheel and tire assemblies front-to-rear. REFER to Section 204-04. If the vehicle drifts/pulls in the same direction, GO to A4 .
No	Tire forces were causing the drift/pull and the concern has been corrected.

A4 CHECK FOR BRAKE DRAG

- Spin all 4 wheel and tire assemblies by hand and check for brake drag.

Do the wheels spin freely?

Yes	GO to A5 .
No	REFER to Section 206-00 to diagnose brake drag condition.

A5 CHECK THE WHEEL ALIGNMENT

- Using alignment equipment and the manufacturer's instructions, check the wheel alignment.

Is the wheel alignment out of specification?

Yes	ADJUST the alignment as necessary. REFER to General Procedures in this section.
No	REFER to Section 211-00A or Section 211-00B to diagnose steering system drift/pull/wander condition.

Component Tests

Ball Joint Inspection

1. Prior to inspecting the ball joints for wear, inspect the wheel bearings. Refer to Section 100-04.
2. **NOTE:** *In order to obtain accurate measurements, the suspension must be in full rebound with the weight of the vehicle supported by the frame.*

Raise and support the vehicle by the frame to allow the wheels to hang in the rebound position.

3. Inspect the ball joint and ball joint boot for damage.
 - If the ball joint or ball joint boot is damaged, install a new ball joint as necessary. Refer to Section 204-01A for Rear Wheel Drive (RWD) vehicles and Section 204-01B for Four-Wheel Drive (4WD) vehicles.

NOTE: Carry out Steps 4 through 6 to inspect the lower ball joint. Carry out Steps 7 through 9 to inspect the upper ball joint.

4. **NOTICE:** Do not use any tools or equipment to move the wheel and tire assembly or suspension components while checking for relative movement. Suspension damage can occur. The use of tools or equipment will also create relative movement that may not exist when using hand force. Relative movement must be measured using hand force only.

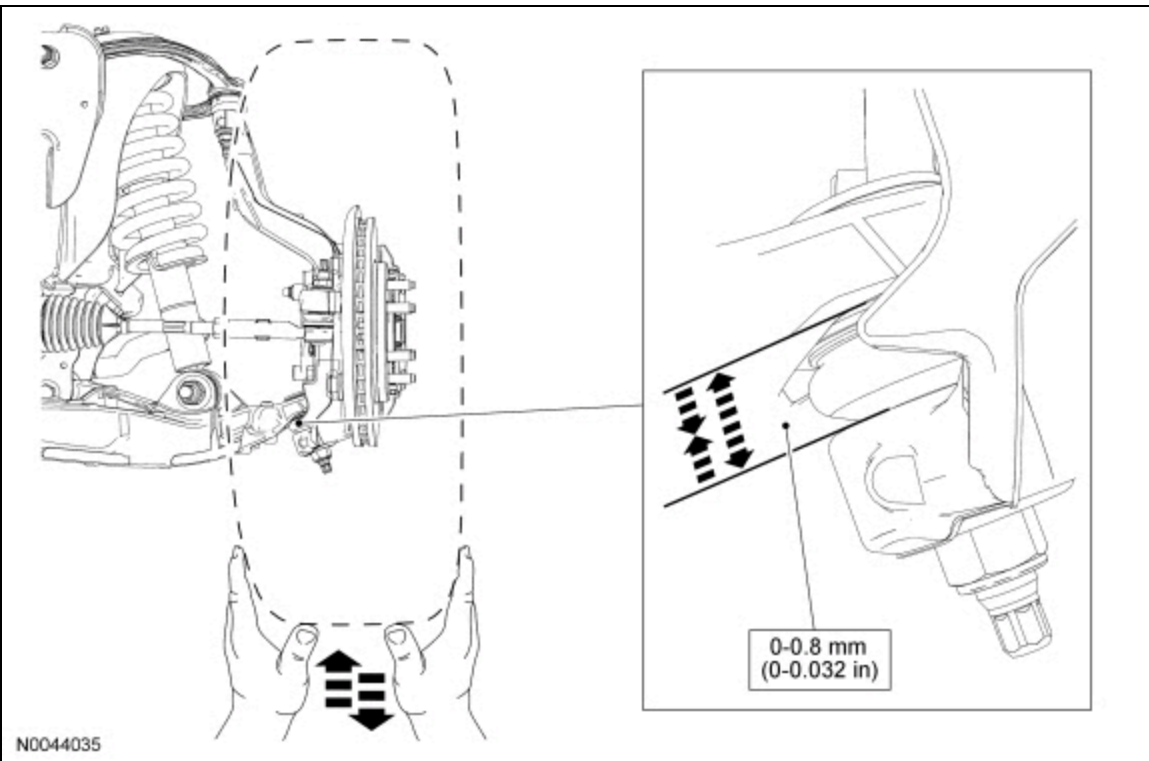
NOTE: The weight of the wheel and tire assembly must be overcome to obtain an accurate measurement on the dial indicator.

Inspect the ball joint for relative movement by alternately pulling downward and pushing upward on the wheel and tire assembly by hand. Note any relative vertical movement between the wheel knuckle and lower arm at the lower ball joint.

- If relative movement is not felt or seen, the ball joint is OK. Do not install a new ball joint.
- If relative movement is found, continue with Step 5.

5. **NOTE:** In order to obtain an accurate measurement, the dial indicator should be aligned as close as possible with the vertical axis (center line) of the ball joint.

To measure ball joint deflection, attach a suitable dial indicator with a flexible arm between the lower control arm and the wheel knuckle or ball joint stud.



6. Measure the ball joint deflection while an assistant pushes up and pulls down on the wheel and tire assembly, by hand.
- If the deflection exceeds the specification, a new ball joint must be installed. Refer to Section 204-01A for RWD vehicles and Section 204-01B for 4WD vehicles.
 - If the deflection meets or is below the specification, continue with the procedure.

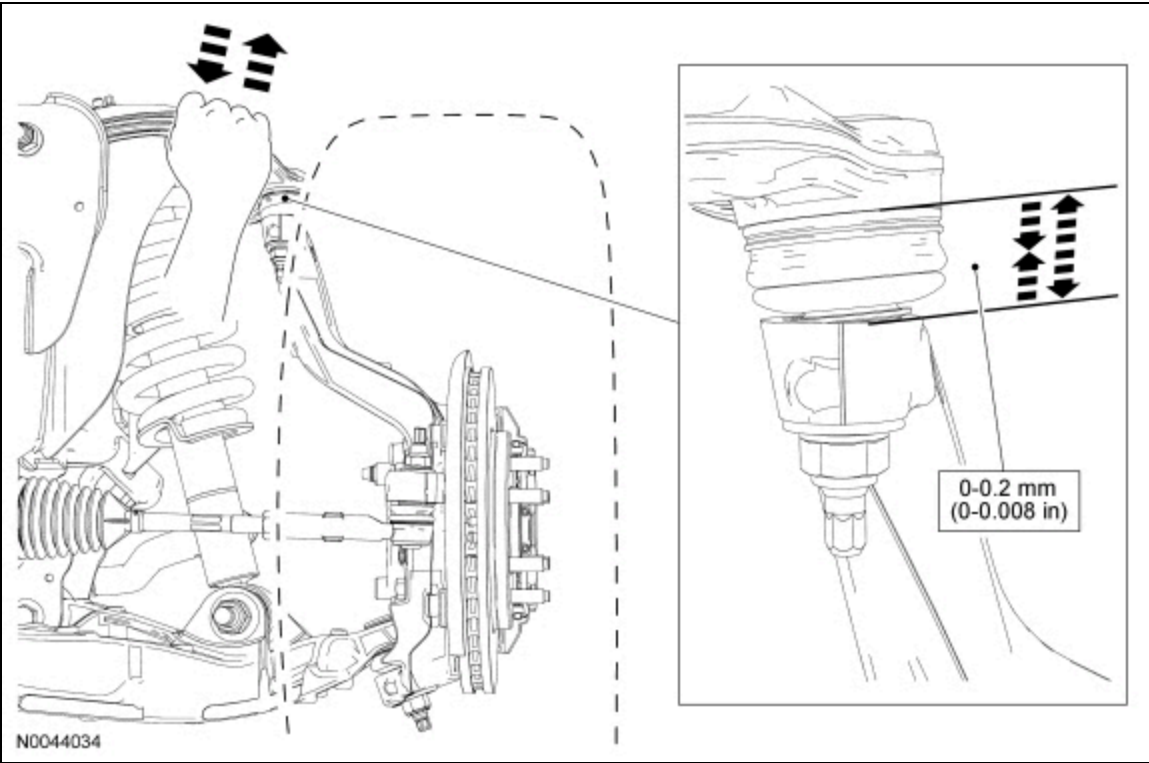
7. **NOTICE:** Do not use any tools or equipment to move the wheel and tire assembly or suspension components while checking for relative movement. Suspension damage can occur. The use of tools or equipment will also create relative movement that may not exist when using hand force. Relative movement must be measured using hand force only.

Inspect the ball joint for relative movement by alternately pulling downward and pushing upward on the upper control arm by hand. Note any relative vertical movement between the wheel knuckle and upper arm at the upper ball joint.

- If relative movement is not felt or seen, the ball joint is OK. Do not install a new ball joint.
- If relative movement is found, continue with Step 8.

8. **NOTE:** *In order to obtain an accurate measurement, the dial indicator should be aligned as close as possible with the vertical axis of the ball joint.*

To measure ball joint deflection, attach a suitable dial indicator with a flexible arm between the upper control arm and the wheel knuckle or ball joint stud.



9. Measure the ball joint deflection while an assistant pushes up and then pulls down on the upper control arm, by hand.
- If the deflection exceeds the specification, a new ball joint must be installed. Refer to Section 204-01A for RWD vehicles and Section 204-01B for 4WD vehicles.
 - If the deflection meets or is below the specification, no further action is required.